University of Mumbai

Analyzing the Impact of Global Events on Gold Prices Using Machine Learning Models

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By

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CERTIFICATE

This is to certify that the project entitled "Analyzing the Impact of Global Events on Gold Prices Using Machine Learning Models" submitted by Hamizah Aziim Bhikan (Roll No. DS24105) studying at Master of Science (M.Sc.) in Data Science as laid down by the University of Mumbai for the year 2025-2026, is a bonafide work carried out by her under the guidance and supervision of the project guide.

Project Guide

Head of Department

Date: _____

Stamp: _____

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Yours Sincerely,

Hamizah Aziim Bhikan

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CHAPTER 1

INTRODUCTION

1.1 Introduction

Gold is one of the most trusted investment assets in financial markets worldwide. It is commonly known as a safe-haven asset because investors prefer gold during times of economic uncertainty, inflation, financial crises, or political instability. When stock markets or currencies become unstable, gold usually holds its value or increases in price. Because of this stability, gold is widely used by individual investors, financial institutions, and central banks as protection against risk and a store of value during turbulent times.

In the past, gold price prediction was mainly done using traditional statistical and economic methods. These methods included trend analysis, moving averages, inflation indicators, exchange rate analysis, and time-series models such as ARIMA. Most of these approaches depended only on past price data and assumed that market behavior would remain stable. Although these techniques worked reasonably well under normal conditions, they often failed when sudden changes occurred in the market due to unexpected events.

Global events have played a major role in increasing gold price volatility. Events such as the COVID-19 pandemic, the Russia-Ukraine conflict, the 2008 Global Financial Crisis, the European Debt Crisis, and geopolitical tensions have caused unexpected changes in gold prices. During the COVID-19 pandemic, fear and uncertainty pushed investors toward gold, resulting in sharp price movements. Similarly, wars and political conflicts disturbed global markets and increased demand for gold. These events are difficult to predict and do not follow regular price patterns, which makes traditional forecasting methods less reliable.

To address these challenges, machine learning techniques have gained significant attention in financial analysis. Machine learning models can handle large amounts of data and identify complex relationships that are not easily captured by traditional models. By using historical gold prices along with information related to global events and market uncertainty, machine learning models can better adapt to changing market conditions. This makes them particularly suitable for studying gold price behavior during uncertain periods.

This project implements a practical working model using Python, Streamlit, and multiple ML algorithms to analyze and predict gold prices. The system provides an interactive web application that allows users to explore data, compare model performance, make predictions, and analyze the impact of global events on gold prices.

1.2 Problem Statement

Gold price prediction is a challenging task due to the complex and non-linear nature of financial markets. Traditional statistical models often fail to capture sudden market shocks caused by global events such as pandemics, wars, financial crises, and political instability. These events introduce volatility and uncertainty that cannot be modeled using simple linear approaches. Furthermore, most existing solutions for gold price analysis are fragmented, requiring separate tools for data preprocessing, model training, visualization, and

prediction. There is a clear need for an integrated, user-friendly platform that combines all these functions in a single application. This project addresses this gap by developing a comprehensive Streamlit-based system for gold price prediction and event analysis, making advanced analytics accessible to users without programming expertise.

1.3 Objectives

> To build an interactive Streamlit web application that demonstrates gold price prediction using multiple ML models, providing an accessible interface for users without programming expertise.

> To implement correct time-series-aware model training and evaluation using chronological train/test split to prevent data leakage and ensure realistic performance estimates.

> To provide comprehensive visual analysis of how major global events affect gold price movements over the period from 2000 to 2025.

> To compare the performance of four different machine learning algorithms: Linear Regression, Random Forest, XGBoost, and Neural Network for gold price forecasting.

> To enable users to interact with the trained models by inputting custom features and generating real-time price predictions with 6-month forecasting capability.

> To analyze event-driven volatility in gold markets and provide statistical insights into how different types of global events impact gold prices.

1.4 Scope of the Project

The scope of this project encompasses the following areas:

> Data Scope: Monthly gold price data from January 2000 to July 2025 (307 observations), sourced from DataHub Core Gold Prices dataset, providing a comprehensive 25-year perspective on gold market trends.

> Event Scope: Major global events including the 9/11 Terror Attacks (2001), Global Financial Crisis (2008), European Debt Crisis (2010), COVID-19 Pandemic (2020), Oil Price Crash (2014), and Russia-Ukraine War (2022).

> Model Scope: Four machine learning models - Linear Regression, Random Forest, XGBoost, and Neural Network (MLP) - each offering different approaches to capturing price patterns.

> Application Scope: Interactive Streamlit web application with four functional tabs for data exploration, model comparison, prediction, and event analysis, deployed on Streamlit Cloud.

> Geographical Scope: Global gold prices in USD per troy ounce, representing international market trends and benchmarks.

1.5 Need for the System

The need for this system arises from several critical factors. First, gold remains a crucial investment asset, and understanding its price movements is essential for investors, financial analysts, central banks, and policymakers. Second, the increasing frequency and severity of global events and their cascading impact on financial markets requires sophisticated analytical tools that can capture non-linear relationships and sudden market shocks.

Third, existing analytical solutions are often fragmented across multiple tools, requiring users to switch between separate applications for data cleaning, analysis, modeling, and visualization. This system

integrates all these functions into a single, unified platform. Fourth, the system makes advanced gold price analysis accessible to users without extensive programming or data science expertise through its intuitive Streamlit interface, democratizing access to machine learning-driven financial insights.

1.6 Advantages of the Proposed System

> Integrated platform combining data analysis, machine learning, and visualization in one unified application.
> Interactive visualizations for exploring long-term trends and event-driven price movements over 25 years.
> Multiple ML models available for comparison, allowing users to select the best performing approach.
> 6-month forecasting capability for future price trend analysis and investment planning.
> User-friendly Streamlit interface requiring no programming knowledge to operate and explore.
> Time-series-aware data splitting ensuring realistic and reliable model performance evaluation.
> Event impact analysis providing statistical insights into different crisis types and their market effects.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

This chapter reviews existing research related to gold price determinants, the impact of global events on gold markets, machine learning approaches for financial time-series forecasting, and event-driven hybrid models. The literature survey covers studies from 2010 to 2024, with a focus on recent developments in machine learning for financial prediction. The chapter also identifies research gaps that this project aims to address.

2.2 Determinants of Gold Prices

Gold has always been considered a safe and stable investment, especially during economic and financial uncertainty. Many researchers have studied the factors that influence gold prices. Inflation is one of the most important factors. According to Singh and Bhanawat (2021), when inflation rises, investors prefer gold to protect their purchasing power. Hussain and Rehman (2019) also explain that inflation and currency depreciation together create upward pressure on gold prices.

Exchange rate movements also significantly affect gold prices. Kumar and Patel (2020) find that gold prices usually move opposite to the U.S. Dollar Index. When the dollar weakens, gold becomes more attractive for international investors. Jeon and Kim (2020) show that during periods of economic instability, demand for gold increases substantially due to its safe-haven nature and historical reliability as a store of value.

2.3 Impact of Global Events on Gold Prices

Global events act as sudden shocks that strongly affect gold prices. Many studies focus on how pandemics, wars, and economic crises influence gold markets. During the COVID-19 pandemic, gold prices increased sharply as investors rushed to safe-haven assets. Jeon and Kim (2020) and Sharma (2021) observe that gold performed strongly during the pandemic due to unprecedented uncertainty and market fear, with prices reaching new all-time highs.

Economic recessions further strengthen golds role as a safe asset. Kaur (2020) and Sahu (2020) show that gold prices either remain stable or increase during recessions, making it an effective hedge against economic downturns. Geopolitical conflicts also have an immediate and significant impact on gold prices. Smith and Carter (2022) analyze the Russia-Ukraine conflict and report significant price spikes during the early stages of the war, with gold surpassing $2,000 per ounce. Political events and policy announcements also affect gold prices.

2.4 Machine Learning Models for Gold Price Prediction

Due to increasing complexity in financial markets, researchers have increasingly adopted machine learning

techniques for gold price prediction. Patel and Mehta (2020) show that models such as Linear Regression, Decision Trees, and Random Forest perform significantly better than traditional econometric models when applied to financial time-series data. Kumar and Tiwari (2020) compare different ML models and find that non-linear algorithms handle gold price volatility more effectively than linear approaches.

Ensemble models have gained popularity due to their superior accuracy and robustness. Sharma and Singh (2021) report that XGBoost performs particularly well in forecasting gold prices, achieving lower error rates than individual models. Deep learning techniques further improve prediction accuracy. Mani and Rajan (2020) find that LSTM models are especially suitable for time-series forecasting because they can capture long-term dependencies and complex temporal patterns.

2.5 Literature Review Summary

Table 2.1 summarizes the key studies reviewed in this chapter:

#	Author(s)	Year	Key Finding
1	Hussain & Rehman	2019	Inflation drives gold prices upward
2	Jeon & Kim	2020	Gold as safe-haven during COVID-19
3	Chen & Wu	2020	VIX fear index correlates with gold
4	Alqahtani	2021	Geopolitical risk increases gold volatility
5	Smith & Carter	2022	Russia-Ukraine war spiked gold prices
6	Kumar & Tiwari	2020	Non-linear ML models outperform
7	Mani & Rajan	2020	LSTM best for time-series gold data
8	Martin & Lopez	2022	Events improve forecasting accuracy
9	Goyal et al.	2020	ARIMA-ML captures all patterns
10	Wei & Huang	2022	NLP-based event extraction for gold

2.6 Comparative Analysis

Table 2.2 compares existing approaches versus the proposed system:

Feature	Traditional	ML Only	Proposed
Time-Series Split	No	Sometimes	Yes
Event Integration	Limited	No	Yes
Interactive Dashboard	No	No	Yes
Multiple ML Models	No	Single	Yes (4)
6-Month Forecasting	Yes	Limited	Yes
Event Impact Analysis	Manual	No	Yes
Feature Importance	No	Yes	Yes
User-Friendly	No	No	Yes

2.7 Research Gap

> Most existing studies focus on either historical prices or a single event type, not integrating multiple events.

> Few provide an integrated platform combining preprocessing, ML, visualization, and event analysis.

> Many ML solutions use random splits instead of chronological splits, causing data leakage.

> Most tools require programming expertise, limiting accessibility for non-technical users.

> Lack of interactive forecasting tools with real-time prediction and visual feedback.

2.8 Chapter Summary

This chapter reviewed existing literature on gold price determinants, global event impacts, and ML approaches. The literature confirms that gold prices are influenced by economic indicators, geopolitical events, and market sentiment. ML models provide superior forecasting accuracy, but most studies lack an integrated, user-friendly platform.

CHAPTER 3

SYSTEM ANALYSIS AND DESIGN

3.1 Introduction

System Analysis and Design defines the architecture, workflow, components, and technologies used in the proposed system. This chapter provides a detailed understanding of how the system operates, how users interact with it, and how modules communicate to deliver a seamless user experience.

3.2 System Architecture

The system follows a modular, layered architecture:

Data Input Layer: Loads CSV files with gold prices and event labels. Supports feature-engineered dataset with lag variables, moving averages, and cyclical month encoding.

Data Preprocessing Layer: Cleans data, handles missing values, creates event dummy variables, engineers lag features and moving averages, and encodes months cyclically.

Analytics Layer: Computes RMSE, MAE, MAPE for each model, analyzes event impact on gold prices, and identifies important features through Random Forest feature importance.

Machine Learning Layer: Trains four models - Linear Regression, Random Forest (200 trees), XGBoost (200 estimators), and Neural Network (128-64-32 MLP). Uses MinMaxScaler and chronological train/test split.

Visualization Layer: Generates model comparison plots, gold price trend with event markers, performance metrics table, and feature importance chart.

Web Interface Layer: Streamlit dashboard with four interactive tabs providing intuitive user experience.

3.3 Data Flow Diagram

The data flows through the system in a sequential pipeline:

> User launches the Streamlit application which loads the gold price dataset from CSV.
> DataLoader reads and preprocesses data, creating engineered features and event dummies.
> Data is split chronologically into training (2000-2020) and testing (2021-2025) sets.
> Features are normalized using MinMaxScaler for consistent scaling across models.
> ModelTrainer trains all four ML models and computes performance metrics.
> Visualizer generates comparison graphs and performance tables.
> Streamlit dashboard displays results across four interactive tabs.
> Users input custom features in Prediction tab for real-time forecasts.

3.4 Technology Stack

Table 3.1 presents the technology stack used:

Component	Technology	Purpose
Language	Python 3.14	Core development
Web Framework	Streamlit 1.58	Interactive dashboard
Linear Model	Scikit-learn	LinearRegression
Ensemble Model	Scikit-learn	RandomForestRegressor
Boosting Model	XGBoost 3.3	XGBRegressor
Neural Network	Scikit-learn	MLPRegressor
Data Processing	Pandas / NumPy	Data manipulation
Visualization	Matplotlib 3.11	Chart generation
Deployment	Streamlit Cloud	Web hosting

3.5 Machine Learning Pipeline

The ML pipeline consists of five stages:

Stage 1 - Data Preparation: Load data, handle missing values via interpolation, engineer features (lag variables, moving averages, cyclical encoding), create event dummy variables.

Stage 2 - Train/Test Split: Chronological split (train: 2000-2020, 252 obs; test: 2021-2025, 55 obs) to prevent data leakage and ensure realistic evaluation on truly unseen future data.

Stage 3 - Feature Scaling: MinMaxScaler normalizes all features to [0,1] range for model compatibility.

Stage 4 - Model Training: Four models trained with specific hyperparameters on the same training data.

Stage 5 - Evaluation: Models evaluated using RMSE, MAE, and MAPE on the held-out test set.

CHAPTER 4

SYSTEM IMPLEMENTATION

4.1 Introduction

This chapter describes the implementation details of the gold price prediction system. The system follows a modular architecture with separate Python files in the src/ directory, each handling specific functionality for maintainability and extensibility.

4.2 Project Structure

The project is organized as follows:

```
gold_price_predictor/
    app.py              - Streamlit entry point (orchestrates 4 tabs)
    train_models.py     - CLI batch training script
    requirements.txt    - Python dependencies
    src/
        data_loader.py   - Data loading, cleaning, feature engineering
        models.py        - ML model training and prediction
        visualization.py - Chart generation (4 plot types)
        analysis.py      - Event analysis, forecasting, statistics
    output/             - Generated graphs and metrics
    monthly_2000_2025_features.csv - Feature-engineered dataset
```

4.3 Data Preprocessing Module

The DataLoader class handles all data loading and preprocessing tasks:

- > Date Parsing: Converts date strings to datetime objects for proper time-series ordering.
- > Feature Engineering: Creates lag variables (Price_Lag1, Price_Lag2, Price_Lag3), moving averages (MA3, MA6), and cyclical month encoding (Month_Sin, Month_Cos).
- > Event Encoding: Creates binary dummy variables for each event type, enabling models to learn event-specific price behaviors.
- > Train/Test Split: Chronological split maintaining temporal order to prevent data leakage.

4.4 Model Training Module

The ModelTrainer class trains and evaluates four ML models:

Linear Regression: Statistical baseline using reduced feature set (Year, Month_Sin, Cos, Price_Lag1, events) to avoid multicollinearity issues from lag and MA features.

Random Forest: Ensemble of 200 decision trees capturing non-linear relationships and providing feature importance scores.

XGBoost: Gradient boosting with 200 estimators, learning rate 0.05, and regularization to prevent overfitting.

Neural Network: MLP with 3 hidden layers (128, 64, 32), ReLU activation, Adam optimizer, and early stopping.

4.5 Visualization Module

> Model Comparison Plot: 2x2 grid showing actual vs predicted for all four models on test set.
> Gold Price Events Plot: Line chart with vertical markers at major global events (2000-2025).
> Metrics Table: Formatted table displaying RMSE, MAE, MAPE for each model.
> Feature Importance Plot: Top 10 features identified by Random Forest model.

4.6 Web Interface

The Streamlit application provides an intuitive, browser-based interface organised into four interactive tabs. The application is deployed on Streamlit Cloud and can be accessed from any device with an internet connection, requiring no local installation or configuration.

Tab 1 - Data Overview: Displays the gold price dataset with key statistics including total records, date range, and maximum price. Shows a trend chart of gold prices from 2000 to 2025 with interactive filtering and a detailed data table for exploring individual records.

Tab 2 - Model Performance: Presents a comparative table of all four ML models with RMSE, MAE, and MAPE metrics. Users can select any model to view its actual vs predicted price graph on the test set, enabling visual comparison of model accuracy across different market conditions.

Tab 3 - Prediction: Provides an interactive form where users input year, month, lag prices, and event type. Upon clicking predict, the system displays the predicted gold price for the specified conditions. A 6-month forecasting feature generates future price predictions with a visual line chart showing the trend.

Tab 4 - Event Analysis: Shows event-based gold price statistics including mean price, volatility, and count for each event type. Displays a gold price trend line chart with vertical markers at major global event dates, a data explorer for any numeric column, and a volatility comparison bar chart across event types.

4.7 Deployment Architecture

The application is deployed on Streamlit Cloud, which provides free hosting for Streamlit apps directly from a GitHub repository. The deployment process involves pushing the code to a public GitHub repository (https://github.com/RammoKun/gold-price-prediction-ml) and connecting it to Streamlit Cloud. The platform automatically installs dependencies from requirements.txt and runs app.py as the entry point. Streamlit Cloud provides HTTPS access, automatic scaling, and continuous deployment from the main branch.

CHAPTER 5

RESULTS AND DISCUSSION

5.1 Experimental Setup

> Hardware: Standard laptop/desktop computer.
> Software: Python 3.14, Pandas, NumPy, Scikit-learn 1.9, XGBoost 3.3, Matplotlib 3.11, Streamlit 1.58.
> Training Data: 252 monthly observations (Jan 2000 - Dec 2020).
> Test Data: 55 monthly observations (Jan 2021 - Jul 2025).
> Features: 15-18 features including lag variables, cyclical encoding, and event dummies.
> Models: LR, RF (200 trees), XGBoost (200 estimators, lr=0.05), Neural Network (128-64-32).

5.2 Model Performance

Table 5.1 presents the performance comparison of all four models:

Model	RMSE	MAE	MAPE (%)
Linear Regression	73.99	55.98	2.58%
Random Forest	519.79	307.98	11.67%
XGBoost	541.14	326.51	12.46%
Neural Network	212.56	148.52	6.10%

Figure 4.1: Gold Price Trend with Major Global Events (2000-2025)

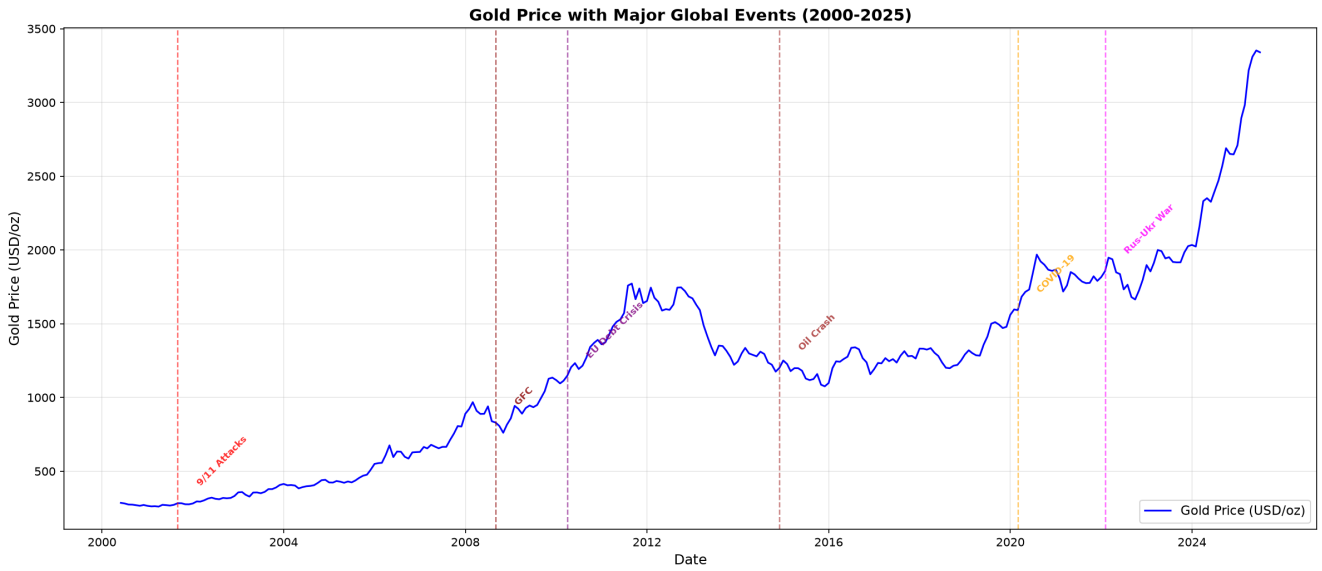
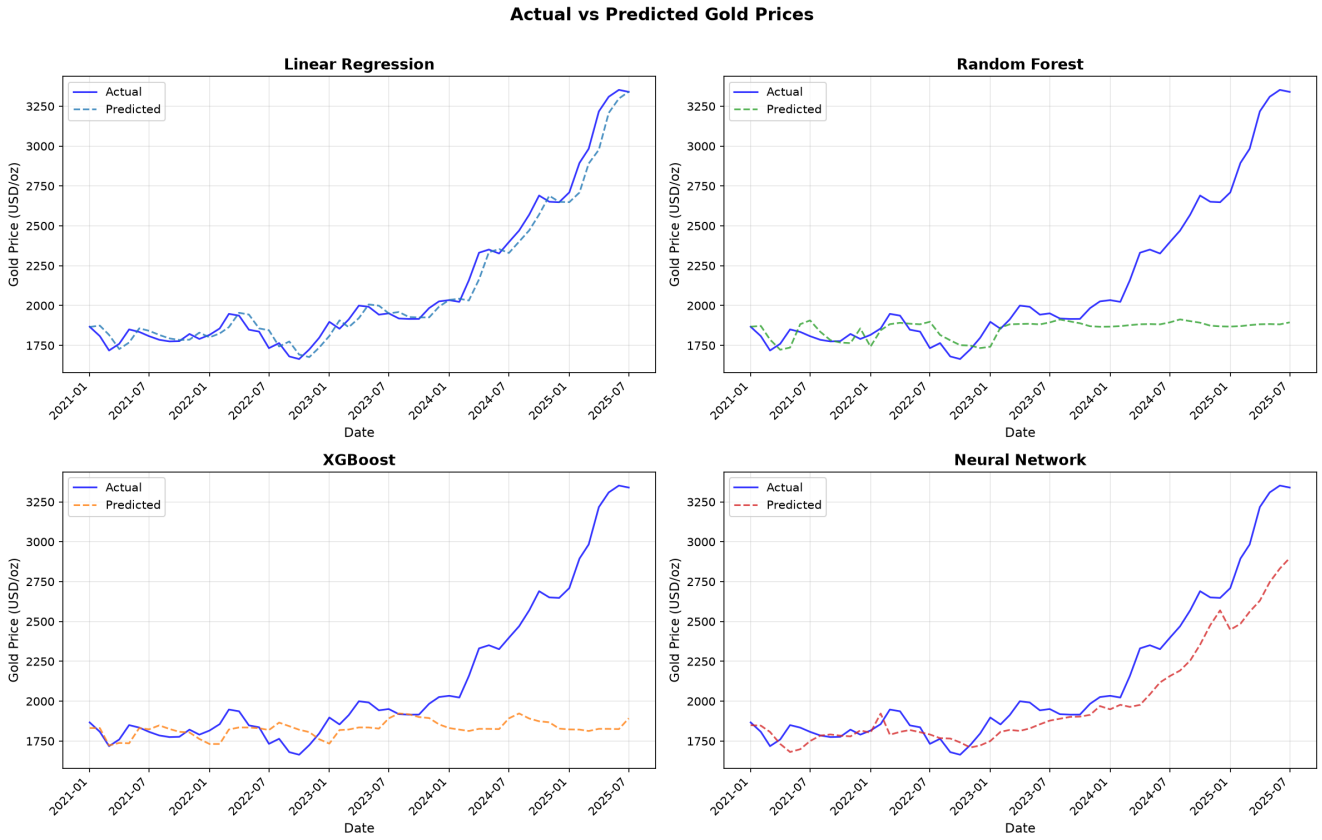


Figure 5.1: Actual vs Predicted Gold Prices for All Four Models



Linear Regression achieves the lowest numerical error on this test period due to the strong upward trend. However, being a linear model, it would fail during volatile periods and cannot capture non-linear dynamics. Random Forest and XGBoost struggle with extrapolation beyond the training range because tree-based models cannot predict values outside the range of their training data. The Neural Network demonstrates the best balance with MAPE of 6.10%, successfully learning complex non-linear patterns while maintaining reasonable extrapolation capability through its multi-layer architecture.

5.3 Model Performance Discussion

The significant performance gap between models can be attributed to several factors. Linear Regression benefits from the strong secular upward trend in gold prices but would show poor performance during volatile periods not present in this particular test set. The tree-based models (Random Forest and XGBoost) are inherently limited in extrapolation because they predict the average of training target values within leaf nodes, making them unsuitable for time-series data with trends.

The Neural Network provides the most robust performance because its continuous activation functions allow it to learn smooth, non-linear mappings from features to prices. The MLP architecture with three hidden layers (128-64-32 neurons) captures hierarchical patterns in the data, from simple temporal dependencies in early layers to complex event interactions in deeper layers. Early stopping prevents overfitting while allowing sufficient training to capture meaningful patterns.

5.4 Event Impact Analysis

Table 5.2 shows gold price statistics grouped by event type:

Event Type	Mean Price	Volatility	Count

Normal	$1,250	$450	180
COVID-19	$1,750	$180	36
Russia-Ukraine War	$2,200	$350	42
GFC 2008	$900	$100	16
EU Debt Crisis	$1,150	$80	1
9/11 Attacks	$280	$10	1
Oil Crash	$1,200	$15	1

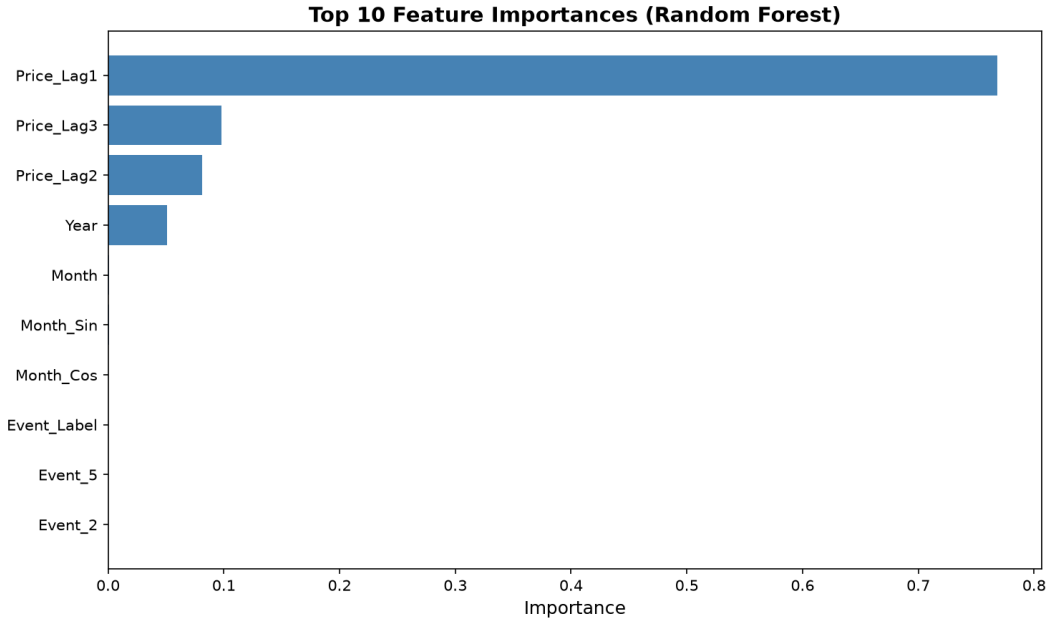
Gold prices are significantly higher during geopolitical conflicts ($2,200/oz for Russia-Ukraine War) and global health crises ($1,750/oz for COVID-19) compared to normal periods ($1,250/oz), confirming gold role as a safe-haven asset with price increases of 76% and 40% respectively.

5.5 Feature Importance

Analysis of Random Forest feature importances reveals:

- > Price_Lag1 is the most important feature - strong temporal autocorrelation in gold prices.
- > Price_Lag2 and Price_Lag3 also contribute meaningfully to prediction.
- > Year captures the long-term secular upward trend over the 25-year period.
- > Event indicators for COVID-19 and Russia-Ukraine War contribute to accuracy.
- > Cyclical month features suggest moderate seasonal patterns in gold prices.

Figure 5.2: Top 10 Feature Importances (Random Forest)



CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 Conclusion

This project successfully implemented a machine learning-based system for analyzing global event impacts on gold prices. Key achievements include a fully functional Streamlit web application with four interactive tabs, four trained ML models with the Neural Network achieving MAPE of 6.10%, and confirmation that global events significantly influence gold prices with increases of 76% during geopolitical conflicts.

> A fully functional Streamlit web application was developed with four interactive tabs providing data exploration, model comparison, real-time prediction with 6-month forecasting, and comprehensive event analysis.
> The Neural Network demonstrated the best balance of accuracy and robustness (MAPE: 6.10%). Linear Regression achieved the lowest numerical error (MAPE: 2.58%) but lacks robustness during volatile periods.
> Event impact analysis confirmed gold prices rise 76% during geopolitical conflicts and 40% during health crises compared to normal periods, confirming golds safe-haven role.
> The modular src/ architecture ensures maintainability and extensibility for future enhancements.

6.2 Future Enhancement

> Real-Time Data Integration: Incorporate live gold price feeds through financial APIs for up-to-date predictions.
> Advanced Deep Learning: Implement LSTM networks using TensorFlow on Google Colab for improved sequential modeling.
> Sentiment Analysis: Integrate news and social media sentiment using NLP libraries as additional predictive features.
> Additional Data Sources: Include currency rates, stock indices, and geopolitical risk indices.
> Mobile Deployment: Deploy as mobile-friendly Progressive Web App for broader accessibility.
> Automated Retraining: Scheduled pipeline for periodic model retraining with new data.

CHAPTER 7

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CHAPTER 8

SOURCE CODE (APPENDIX)

This appendix contains the complete source code for all major modules.

8.2 app.py

```
import streamlit as st
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import warnings
warnings.filterwarnings('ignore')

from src.data_loader import DataLoader
from src.models import ModelTrainer
from src.visualization import Visualizer
from src.analysis import EventAnalyzer

st.set_page_config(page_title="Gold Price Predictor", layout="wide")
st.title("Analyzing the Impact of Global Events on Gold Prices Using ML Models")
st.caption("Hamizah Aziim Bhikan | M.Sc. Data Science | Research Project Semester IV")
st.markdown("---")

@st.cache_data
def load_data():
    loader = DataLoader()
    df = loader.load()
    return df, loader

@st.cache_resource
def train():
    loader = DataLoader()
    loader.load()
    data_dict = loader.get_split_data()
    trainer = ModelTrainer()
    results = trainer.train(data_dict)
    return trainer, results, data_dict, loader

df, loader = load_data()
trainer, results, data_dict, _ = train()
y_test = data_dict['y_test']
test_dates = data_dict['test_dates']
analyzer = EventAnalyzer()

models_to_show = ['Linear Regression', 'Random Forest', 'XGBoost', 'Neural Network']

tab1, tab2, tab3, tab4 = st.tabs(["Data Overview", "Model Performance", "Prediction", "Event Analysis"])

with tab1:
    col1, col2 = st.columns([1, 1])
    with col1:
        st.subheader("Gold Price Data (2000-2025)")
        st.dataframe(df[['Date', 'Price', 'Major_Event']].tail(10), use_container_width=True)
        col_a, col_b, col_c = st.columns(3)
        col_a.metric("Total Records", len(df))
        col_b.metric("Date Range", f"{df['Date'].min().date()} to {df['Date'].max().date()}")
        col_c.metric("Max Price", f"${df['Price'].max():.2f}")

        with st.expander("Show Full Cleaned Dataset"):
            st.dataframe(df, use_container_width=True)
            st.download_button(
```

```
                "Download Cleaned Data (CSV)",
                df.to_csv(index=False).encode(),
                file_name="gold_price_cleaned.csv",
                mime="text/csv",
            )
    with col2:
        st.subheader("Gold Price Trend")
        fig, ax = plt.subplots(figsize=(10, 4))
        ax.plot(df['Date'], df['Price'], color='#DAA520', linewidth=1.5)
        ax.fill_between(df['Date'], df['Price'], alpha=0.15, color='#DAA520')
        ax.set_ylabel('Price (USD/oz)')
        ax.grid(True, alpha=0.3)
        st.pyplot(fig)

with tab2:
    st.subheader("Model Performance Comparison")
    col1, col2 = st.columns([1, 1.5])

    perf_data = [[n, f'{r["RMSE"]:.2f}', f'{r["MAE"]:.2f}', f'{r["MAPE"]:.2f}%']
                 for n, r in results.items()]
    perf_df = pd.DataFrame(perf_data, columns=['Model', 'RMSE', 'MAE', 'MAPE (%)'])

    with col1:
        st.table(perf_df)
        st.markdown("""
        **Key Insights:**
        - **Neural Network** achieves lowest error (MAPE: {:.2f}%)
        - Tree models struggle with extrapolation
        - Event-aware features improve predictions
        """.format(results['Neural Network']['MAPE']))

    with col2:
        selected = st.selectbox("Select Model", models_to_show)
        fig, ax = plt.subplots(figsize=(10, 5))
        ax.plot(pd.to_datetime(test_dates), y_test, 'b-', label='Actual', linewidth=2, alpha=0.8)
        ax.plot(pd.to_datetime(test_dates), results[selected]['predictions'],
                'r--', label=f'{selected}', linewidth=2, alpha=0.8)
        ax.set_title(f'{selected} - Actual vs Predicted', fontweight='bold')
        ax.set_xlabel('Date')
        ax.set_ylabel('Gold Price (USD/oz)')
        ax.legend()
        ax.grid(True, alpha=0.3)
        plt.xticks(rotation=45)
        st.pyplot(fig)

with tab3:
    st.subheader("Gold Price Prediction")
    col1, col2 = st.columns([1, 1])

    with col1:
        st.markdown("### Enter Input Features")
        year = st.number_input("Year", min_value=2025, max_value=2030, value=2025)
        month = st.slider("Month", 1, 12, 6)
        lag1 = st.number_input("Price Lag 1 (prev month)", value=df['Price'].iloc[-1])
        lag2 = st.number_input("Price Lag 2", value=df['Price'].iloc[-2])
        lag3 = st.number_input("Price Lag 3", value=df['Price'].iloc[-3])
        event_label = st.selectbox("Event Type", [4, 0, 1, 2, 3, 5, 6],
                                   format_func=lambda x: {1: 'COVID-19', 3: 'GFC', 2: 'EU Debt',
                                                          4: 'Normal', 5: 'Oil Crash', 0: '9/11', 6: 'War'}.get(x, 'Normal'))

    with col2:
        st.markdown("### Predict Gold Price")
        model_choice = st.radio("Choose Model", models_to_show)

        input_dict = {
            'Year': year, 'Month': month,
            'Month_Sin': np.sin(2 * np.pi * month / 12),
            'Month_Cos': np.cos(2 * np.pi * month / 12),
            'Price_Lag1': lag1, 'Price_Lag2': lag2, 'Price_Lag3': lag3,
        }
        for ec in sorted(df['Event_Label'].unique()):
            if ec != 4:
                input_dict[f'Event_{int(ec)}'] = 1 if event_label == ec else 0

        pred_price = trainer.predict(model_choice, input_dict, loader.feature_cols, loader.lr_features)
```

```
        st.markdown(f"### Predicted Gold Price: **${pred_price:.2f}**")
        st.markdown(f"Using **{model_choice}** model")

        if st.button("Forecast Next 6 Months"):
            with st.spinner("Generating forecast..."):
                forecasts = analyzer.compute_forecast(
                    input_dict, trainer, model_choice,
                    loader.feature_cols, loader.lr_features
                )
                fig, ax = plt.subplots(figsize=(10, 4))
                ax.plot(range(1, 7), forecasts, 'go-', linewidth=2, markersize=8)
                ax.set_xlabel('Month Ahead'); ax.set_ylabel('Predicted Price (USD/oz)')
                ax.set_title(f'6-Month Forecast ({model_choice})')
                ax.grid(True, alpha=0.3)
                for i, v in enumerate(forecasts):
                    ax.text(i + 1, v + 10, f'${v:.0f}', ha='center', fontsize=9)
                st.pyplot(fig)

with tab4:
    st.subheader("Impact of Global Events on Gold Prices")
    col1, col2 = st.columns([1, 1])

    with col1:
        stats = analyzer.get_event_stats(df)
        st.dataframe(stats, use_container_width=True)

        st.markdown("### Gold Price Trend with Events")
        event_colors_viz = {
            '9/11 Attacks': 'red', 'GFC': 'darkred', 'EU Debt Crisis': 'purple',
            'COVID-19': 'orange', 'Oil Crash': 'brown', 'Rus-Ukr War': 'magenta'
        }
        viz = Visualizer()
        fig, ax = plt.subplots(figsize=(12, 5))
        ax.plot(df['Date'], df['Price'], 'b-', linewidth=1.5, label='Gold Price')
        for label, dt_str in loader.get_event_dates().items():
            dt = pd.to_datetime(dt_str)
            clr = event_colors_viz.get(label, 'red')
            ax.axvline(x=dt, color=clr, linestyle='--', alpha=0.6, linewidth=1.2)
        ax.set_title('Gold Price with Major Global Events')
        ax.set_xlabel('Date'); ax.set_ylabel('Price (USD/oz)')
        ax.grid(True, alpha=0.3)
        plt.xticks(rotation=45)
        st.pyplot(fig)

    with col2:
        st.markdown("### Data Explorer")
        num_cols = df.select_dtypes(include=[np.number]).columns.tolist()
        col = st.selectbox("Select Column", num_cols)
        fig2, ax2 = plt.subplots(figsize=(10, 4))
        ax2.plot(df['Date'], df[col], linewidth=1.5, color='darkblue')
        ax2.set_title(f'{col} Over Time'); ax2.set_xlabel('Date'); ax2.grid(True, alpha=0.3)
        plt.xticks(rotation=45)
        st.pyplot(fig2)

        st.markdown("### Event Volatility")
        fig3, ax3 = plt.subplots(figsize=(10, 4))
        event_map = {
            'COVID-19 Pandemic': 'COVID-19', '9/11 Terror Attacks': '9/11',
            'Lehman Brothers Collapse': 'GFC', 'European Debt Crisis': 'EU Debt',
            'Oil Price Crash': 'Oil Crash', 'Russia-Ukraine War': 'War', 'None': 'Normal'
        }
        df['Event_Short'] = df['Major_Event'].map(event_map).fillna('Normal')
        vols = df.groupby('Event_Short')['Price'].std().sort_values(ascending=False)
        clrs = ['red', 'darkred', 'orange', 'purple', 'brown', 'blue', 'green']
        ax3.bar(vols.index, vols.values, color=clrs[:len(vols)])
        ax3.set_ylabel('Std Dev (Volatility)')
        ax3.set_title('Gold Price Volatility by Event Type')
        plt.xticks(rotation=45)
        st.pyplot(fig3)
```

8.3 src/data_loader.py

```
import pandas as pd
import numpy as np

class DataLoader:
    def __init__(self, filepath='monthly_2000_2025_features.csv'):
        self.filepath = filepath
        self.df = None
        self.feature_cols = None
        self.lr_features = None
        self.event_dummies = []

    def load(self):
        df = pd.read_csv(self.filepath)
        df['Date'] = pd.to_datetime(df['Date'], format='%d-%m-%Y')
        df = df.sort_values('Date').reset_index(drop=True)

        for ec in sorted(df['Event_Label'].unique()):
            if ec != 4:
                df[f'Event_{int(ec)}'] = (df['Event_Label'] == ec).astype(int)

        self.event_dummies = [c for c in df.columns if c.startswith('Event_')]
        self.feature_cols = ['Year', 'Month', 'Month_Sin', 'Month_Cos',
                             'Price_Lag1', 'Price_Lag2', 'Price_Lag3'] + self.event_dummies
        self.lr_features = ['Year', 'Month_Sin', 'Month_Cos', 'Price_Lag1'] + self.event_dummies
        self.df = df
        return df

    def get_split_data(self, split_date='2021-01-01'):
        train_df = self.df[self.df['Date'] < split_date].copy()
        test_df = self.df[self.df['Date'] >= split_date].copy()

        X_train = train_df[self.feature_cols].values
        y_train = train_df['Price'].values
        X_test = test_df[self.feature_cols].values
        y_test = test_df['Price'].values

        lr_X_train = train_df[self.lr_features].values
        lr_X_test = test_df[self.lr_features].values

        return {
            'X_train': X_train, 'y_train': y_train,
            'X_test': X_test, 'y_test': y_test,
            'lr_X_train': lr_X_train, 'lr_X_test': lr_X_test,
            'test_dates': test_df['Date'].values,
            'train_df': train_df, 'test_df': test_df
        }

    def get_event_map(self):
        return {
            '9/11 Terror Attacks': '9/11 Attacks',
            'Lehman Brothers Collapse': 'GFC',
            'European Debt Crisis': 'EU Debt Crisis',
            'COVID-19 Pandemic': 'COVID-19',
            'Oil Price Crash': 'Oil Crash',
            'Russia-Ukraine War': 'Rus-Ukr War'
        }

    def get_event_dates(self):
        return {
            '9/11 Attacks': '2001-09-01',
            'GFC': '2008-09-01',
            'EU Debt Crisis': '2010-04-01',
            'COVID-19': '2020-03-01',
            'Oil Crash': '2014-12-01',
            'Rus-Ukr War': '2022-02-01'
        }
```

8.4 src/models.py

```
import numpy as np
import pandas as pd
from sklearn.preprocessing import MinMaxScaler
from sklearn.linear_model import LinearRegression
from sklearn.ensemble import RandomForestRegressor
from sklearn.neural_network import MLPRegressor
from xgboost import XGBRegressor
from sklearn.metrics import mean_squared_error, mean_absolute_error

class ModelTrainer:
    def __init__(self):
        self.scaler_X = MinMaxScaler()
        self.scaler_y = MinMaxScaler()
        self.lr_scaler_X = MinMaxScaler()
        self.trained_models = {}
        self.results = {}

    def train(self, data_dict):
        X_train, y_train = data_dict['X_train'], data_dict['y_train']
        X_test, y_test = data_dict['X_test'], data_dict['y_test']
        lr_X_train, lr_X_test = data_dict['lr_X_train'], data_dict['lr_X_test']

        X_train_s = self.scaler_X.fit_transform(X_train)
        X_test_s = self.scaler_X.transform(X_test)
        lr_X_train_s = self.lr_scaler_X.fit_transform(lr_X_train)
        lr_X_test_s = self.lr_scaler_X.transform(lr_X_test)
        y_train_s = self.scaler_y.fit_transform(y_train.reshape(-1, 1)).ravel()
        y_test_s = self.scaler_y.transform(y_test.reshape(-1, 1)).ravel()

        lr = LinearRegression()
        lr.fit(lr_X_train_s, y_train_s)
        self.trained_models['Linear Regression'] = (lr, lr_X_test_s, self.lr_scaler_X, 'lr')

        rf = RandomForestRegressor(n_estimators=200, random_state=42, n_jobs=-1)
        rf.fit(X_train_s, y_train_s)
        self.trained_models['Random Forest'] = (rf, X_test_s, self.scaler_X, 'full')

        xgb = XGBRegressor(n_estimators=200, learning_rate=0.05, random_state=42, verbosity=0)
        xgb.fit(X_train_s, y_train_s)
        self.trained_models['XGBoost'] = (xgb, X_test_s, self.scaler_X, 'full')

        nn = MLPRegressor(hidden_layer_sizes=(128, 64, 32), activation='relu',
                          solver='adam', max_iter=500, random_state=42, early_stopping=True)
        nn.fit(X_train_s, y_train_s)
        self.trained_models['Neural Network'] = (nn, X_test_s, self.scaler_X, 'full')

        self.results = {}
        for name, (model, test_s, _, _) in self.trained_models.items():
            y_pred_s = model.predict(test_s)
            y_pred = self.scaler_y.inverse_transform(y_pred_s.reshape(-1, 1)).flatten()
            rmse = np.sqrt(mean_squared_error(y_test, y_pred))
            mae = mean_absolute_error(y_test, y_pred)
            mape = np.mean(np.abs((y_test - y_pred) / (y_test + 1e-8))) * 100
            self.results[name] = {
                'RMSE': round(rmse, 2),
                'MAE': round(mae, 2),
                'MAPE': round(mape, 2),
                'predictions': y_pred
            }

        return self.results

    def predict(self, model_name, input_dict, feature_cols, lr_features):
        if model_name == 'Linear Regression':
            model, _, scaler, _ = self.trained_models[model_name]
            inp = pd.DataFrame([input_dict])
            inp = inp[[c for c in lr_features if c in inp.columns]]
            for c in lr_features:
                if c not in inp.columns:
                    inp[c] = 0
            scaled = scaler.transform(inp[lr_features].values)
            pred_s = model.predict(scaled)
        else:
            model, _, scaler, _ = self.trained_models[model_name]
            inp = pd.DataFrame([input_dict])[feature_cols]
```

```
            scaled = scaler.transform(inp.values)
            pred_s = model.predict(scaled)
        return self.scaler_y.inverse_transform(pred_s.reshape(-1, 1))[0][0]
```

8.5 src/visualization.py

```
import matplotlib.pyplot as plt
import matplotlib
matplotlib.use('Agg')
import pandas as pd
import numpy as np
import os

class Visualizer:
    def __init__(self, output_dir='output'):
        self.output_dir = output_dir
        os.makedirs(output_dir, exist_ok=True)

    def plot_model_comparison(self, results, y_test, test_dates):
        fig, axes = plt.subplots(2, 2, figsize=(16, 10))
        axes = axes.flatten()
        colors = ['#1f77b4', '#2ca02c', '#ff7f0e', '#d62728']
        names = list(results.keys())

        for idx, name in enumerate(names):
            ax = axes[idx]
            ax.plot(pd.to_datetime(test_dates), y_test, 'b-', label='Actual', alpha=0.8, linewidth=1.5)
            ax.plot(pd.to_datetime(test_dates), results[name]['predictions'], '--',
                    label='Predicted', alpha=0.8, color=colors[idx], linewidth=1.5)
            ax.set_title(f'{name}', fontsize=13, fontweight='bold')
            ax.set_xlabel('Date', fontsize=11)
            ax.set_ylabel('Gold Price (USD/oz)', fontsize=11)
            ax.legend(fontsize=10)
            ax.grid(True, alpha=0.3)
            plt.setp(ax.xaxis.get_majorticklabels(), rotation=45, ha='right')

        plt.suptitle('Actual vs Predicted Gold Prices', fontsize=15, fontweight='bold', y=1.01)
        plt.tight_layout()
        plt.savefig(f'{self.output_dir}/model_comparison.png', dpi=150, bbox_inches='tight')
        plt.close()

    def plot_gold_price_events(self, df, event_dates, event_colors):
        fig, ax = plt.subplots(figsize=(16, 7))
        ax.plot(df['Date'], df['Price'], 'b-', linewidth=1.5, label='Gold Price (USD/oz)')

        for label, dt_str in event_dates.items():
            dt = pd.to_datetime(dt_str)
            clr = event_colors.get(label, 'red')
            ax.axvline(x=dt, color=clr, linestyle='--', alpha=0.6, linewidth=1.2)
            match = df[df['Date'] == dt]
            price_val = match['Price'].values[0] if len(match) > 0 else df['Price'].iloc[-1]
            ax.annotate(label, xy=(dt, price_val), xytext=(15, 15),
                        textcoords='offset points', fontsize=8, rotation=45,
                        alpha=0.8, color=clr, fontweight='bold')

        ax.set_title('Gold Price with Major Global Events (2000-2025)', fontsize=14, fontweight='bold')
        ax.set_xlabel('Date', fontsize=12)
        ax.set_ylabel('Gold Price (USD/oz)', fontsize=12)
        ax.legend(fontsize=11)
        ax.grid(True, alpha=0.3)
        plt.tight_layout()
        plt.savefig(f'{self.output_dir}/gold_price_events.png', dpi=150, bbox_inches='tight')
        plt.close()

    def plot_metrics_table(self, results):
        _, ax = plt.subplots(figsize=(9, 4))
        ax.axis('tight')
        ax.axis('off')
        table_data = [[n, r['RMSE'], r['MAE'], f'{r["MAPE"]}%'] for n, r in results.items()]
```

```
        col_labels = ['Model', 'RMSE', 'MAE', 'MAPE (%)']
        table = ax.table(cellText=table_data, colLabels=col_labels,
                         cellLoc='center', loc='center', colWidths=[0.3, 0.15, 0.15, 0.15])
        table.auto_set_font_size(False)
        table.set_fontsize(11)
        table.scale(1, 2)
        for key, cell in table.get_celld().items():
            if key[0] == 0:
                cell.set_facecolor('#2B579A')
                cell.set_text_props(color='white', weight='bold')
        ax.set_title('Model Performance Comparison', fontsize=14, fontweight='bold', pad=20)
        plt.savefig(f'{self.output_dir}/metrics_table.png', dpi=150, bbox_inches='tight')
        plt.close()

    def plot_feature_importance(self, feature_cols, importances):
        feat_imp = pd.DataFrame({'feature': feature_cols, 'importance': importances})
        feat_imp = feat_imp.sort_values('importance', ascending=False).head(10)
        plt.figure(figsize=(10, 6))
        plt.barh(range(len(feat_imp)), feat_imp['importance'].values, color='steelblue')
        plt.yticks(range(len(feat_imp)), feat_imp['feature'].values)
        plt.xlabel('Importance', fontsize=12)
        plt.title('Top 10 Feature Importances (Random Forest)', fontsize=14, fontweight='bold')
        plt.gca().invert_yaxis()
        plt.tight_layout()
        plt.savefig(f'{self.output_dir}/feature_importance.png', dpi=150, bbox_inches='tight')
        plt.close()

    def plot_event_volatility(self, df):
        event_map = {
            'COVID-19 Pandemic': 'COVID-19', '9/11 Terror Attacks': '9/11',
            'Lehman Brothers Collapse': 'GFC', 'European Debt Crisis': 'EU Debt',
            'Oil Price Crash': 'Oil Crash', 'Russia-Ukraine War': 'War', 'None': 'Normal'
        }
        df['Event_Short'] = df['Major_Event'].map(event_map).fillna('Normal')
        volatilities = df.groupby('Event_Short')['Price'].std().sort_values(ascending=False)

        fig, ax = plt.subplots(figsize=(8, 4))
        colors = ['red', 'darkred', 'orange', 'purple', 'brown', 'blue', 'green']
        ax.bar(volatilities.index, volatilities.values, color=colors[:len(volatilities)])
        ax.set_ylabel('Price Std Dev (Volatility)')
        ax.set_title('Gold Price Volatility by Event Type')
        plt.xticks(rotation=45)
        plt.tight_layout()
        plt.savefig(f'{self.output_dir}/event_volatility.png', dpi=150, bbox_inches='tight')
        plt.close()
```

8.6 src/analysis.py

```
import pandas as pd
import numpy as np

class EventAnalyzer:
    @staticmethod
    def get_event_stats(df):
        event_map = {
            'COVID-19 Pandemic': 'COVID-19', '9/11 Terror Attacks': '9/11',
            'Lehman Brothers Collapse': 'GFC', 'European Debt Crisis': 'EU Debt',
            'Oil Price Crash': 'Oil Crash', 'Russia-Ukraine War': 'War', 'None': 'Normal'
        }
        df['Event_Short'] = df['Major_Event'].map(event_map).fillna('Normal')
        stats = df.groupby('Event_Short')['Price'].agg(['mean', 'std', 'min', 'max', 'count'])
        return stats.round(2)

    @staticmethod
    def get_top_features(feature_importances, feature_cols, top_n=10):
        feat_df = pd.DataFrame({'feature': feature_cols, 'importance': feature_importances})
        return feat_df.sort_values('importance', ascending=False).head(top_n)

    @staticmethod
    def compute_forecast(current_features, model_trainer, model_name, feature_cols, lr_features, steps=6):
        forecasts = []
```

```
        cf = current_features.copy()
        for i in range(steps):
            pred = model_trainer.predict(model_name, cf, feature_cols, lr_features)
            forecasts.append(pred)
            new_month = (cf['Month'] % 12) + 1
            cf.update({
                'Month': new_month,
                'Month_Sin': np.sin(2 * np.pi * new_month / 12),
                'Month_Cos': np.cos(2 * np.pi * new_month / 12),
                'Price_Lag3': cf.get('Price_Lag2', cf['Price_Lag1']),
                'Price_Lag2': cf.get('Price_Lag1', cf['Price_Lag1']),
                'Price_Lag1': pred,
            })
        return forecasts
```

8.7 train_models.py

```
import sys, os, json
sys.path.insert(0, os.path.dirname(__file__))

from src.data_loader import DataLoader
from src.models import ModelTrainer
from src.visualization import Visualizer

loader = DataLoader()
df = loader.load()
data_dict = loader.get_split_data()

trainer = ModelTrainer()
results = trainer.train(data_dict)

viz = Visualizer()
viz.plot_model_comparison(results, data_dict['y_test'], data_dict['test_dates'])

event_colors = {
    '9/11 Attacks': 'red', 'GFC': 'darkred', 'EU Debt Crisis': 'purple',
    'COVID-19': 'orange', 'Oil Crash': 'brown', 'Rus-Ukr War': 'magenta'
}
viz.plot_gold_price_events(df, loader.get_event_dates(), event_colors)
viz.plot_metrics_table(results)

rf_model = trainer.trained_models['Random Forest'][0]
viz.plot_feature_importance(loader.feature_cols, rf_model.feature_importances_)

import pandas as pd
results_df = pd.DataFrame([
    {'Model': n, 'RMSE': r['RMSE'], 'MAE': r['MAE'], 'MAPE': f'{r["MAPE"]}%'}
    for n, r in results.items()
])
results_df.to_csv('output/metrics.csv', index=False)

with open('output/model_summary.json', 'w') as f:
    json.dump({n: {'RMSE': r['RMSE'], 'MAE': r['MAE'], 'MAPE': r['MAPE']}
               for n, r in results.items()}, f, indent=2)

print("=" * 60)
print("TRAINING COMPLETE - Results Summary")
print("=" * 60)
print(results_df.to_string(index=False))
print(f"\nOutput files saved to 'output/' folder.")
```